

A test of higher and lower fractional volumes of resistance training upon arm and thigh muscle area: A multi-site randomised trial*

James Steele^{1,2,3,4}, David Gschneidner⁵, Luke Carlson⁵, James Fisher^{4,6,7}

¹Steele Research Limited, Eastleigh, Hampshire, UK

²MacroFactor, Stronger by Science Technologies LLC, Raleigh, North Carolina, USA

³School of Health and Biomedical Sciences, Royal Melbourne Institute of Technology, Melbourne, VIC, Australia

⁴Department of Sport and Health, Solent University, Southampton, Hampshire UK

⁵Discover Strength, Minneapolis, Minnesota, USA

⁶The Exercise Coach, Tennessee, USA

⁷Department of Rehabilitation & Sport Sciences, Bournemouth University, Bournemouth, Dorset, UK

April 8, 2026

Abstract

Recent work has theorised the effects of resistance training volume to be positive and monotonic, albeit with diminishing returns, with regards to hypertrophy. Improvements in muscle size however are typically small, even smaller in trained people due to the linear-logarithmic adaptation to RT over time, and thus between intervention differences in effects are likely to be very small. As such, in contrast to most studies in the field which aim to detect differences between interventions, we sought to conduct a highly powered pre-registered test of the statistical equivalence of two RT interventions in previously trained participants; namely low (9 fractional sets per week) and high (36 fractional sets per week) volumes. A randomised controlled trial across 22 sites was employed with 125 participants recruited. Our primary outcome was hypertrophy operationalised as estimated muscle cross sectional area using circumference and skinfold measurements of the upper arm and thigh. At the participant level, 120 participants were randomly assigned to either the low ($n = 56$) or high ($n = 64$) volume RT intervention condition. Participants underwent pre-intervention testing and then participated in a 12-week intervention with post-intervention testing following this. Our primary estimand of interest was the condition by time interaction effect from our pre-registered analysis of pooled outcomes reflecting the standardised between condition difference in change in hypertrophy over time. After randomisation 112 participants completed baseline testing and 87 completed post-intervention testing. The estimate for this effect was 0.023 [95%CI: -0.044, 0.091] and the p -value for equivalence was $p=0.032$ supporting statistically equivalent effects between conditions. Main effects for time were also small 0.087 [95%CI: 0.053, 0.121] in line with prior predictions from theoretical linear-log growth models. This study is to our knowledge one of the largest to compare the effects of low and high volume RT interventions upon hypertrophy in previously trained participants. We found statistical equivalence between both conditions and both main effects of time, and any interaction effects for condition by time, are likely small. More broadly, this study further corroborates the linear-log theory of adaptation, that the effects of RT in trained persons should be expected to be small, and that current studies in the field of RT are woefully underpowered to be able to detect their effects, let alone test between intervention comparisons. **Keywords:** resistance training, volume, hypertrophy, trained participants

Introduction

A long-standing, though often debated, theory in resistance training (RT) is that muscular hypertrophy is effected in some dose-response fashion by the volume of RT performed. Indeed, the most recent meta-

*Preprint, please cite as: Steele, J., Gschneidner, D., Carlson, L., and Fisher, J. (2026). A test of higher and lower fractional volumes of resistance training upon arm and thigh muscle area: A multi-site randomised trial. sportrxiv DOI: <https://doi.org/10.51224/SportRxiv.810>. Address for correspondence: james@steele-research.com

regression analyses have estimated, with volume best operationalised as fractional sets (i.e., whereby *indirect* exercises count for half of a set towards volume for that muscle group, and *direct* count as a whole set), a positive monotonic dose-response relationship, albeit with diminishing returns, between both weekly (Pelland et al. 2026), and per session (Remmert et al. 2025), set volumes and muscle hypertrophy. From a practical perspective, these meta-regressions suggest ~11 fractional sets per session with ~31 weekly fractional sets before there is no additionally detectable superiority to greater volumes. This corresponds to a recommendation of ~3 sessions per week performing ~11 sets per session. In terms of the estimated magnitudes, for example, the standardised mean difference (SMD) for a contrast of 9 fractional sets per week against 36 fractional sets per week^a is 0.294 [95% CI: 0.20, 0.39]. Further, the dose-response effect estimated by these meta-regressions does not appear to be moderated by training status which suggests that we should predict, based on this theory, that there are still benefits for higher volumes in trained participants and of a similar comparative magnitude. However, though larger than typical, the meta-regressions supporting this theory of dose-response to RT volume included only k=35 studies on hypertrophy outcomes and the lack of conditional effects by training status combined with contrasts of this magnitude seem unrealistic in light of other background knowledge and theory.

When we contextualise this contrast effect of fractional volume with the findings of a larger meta-analysis (k=111) of varied RT interventions compared to non-training controls in untrained participants, which has estimated the typical effect of RT upon hypertrophy to be only 0.34 [95% confidence interval: 0.29, 0.39] SMD (Steele et al. 2023), the estimated effects for higher volumes seem comparatively large. Further, previous work has demonstrated that RT outcomes respond in a roughly linear-log function with duration of exposure to RT suggesting diminished effects with greater prior training exposure (Steele et al. 2022; Latella et al. 2024)^b. Given this, it seems likely that the simple training effect for any RT intervention over time in previously trained participants for the typical duration (i.e., ~12 weeks) of a study is far less than the 0.34 SMD noted above. In fact, given this theory of linear-log adaptation, the estimated contrast when using the large meta-analysis data set mentioned (Steele et al. 2023) reveals a simple training effect prediction over time of ~ 0.05 SMD for participants with at least 6 months of prior training engaging in a further 12 week intervention duration. Corroborating this theory of linear-log adaptation, and the magnitudes of effects it predicts, the estimates for both the main effect of time, and time-by-condition interaction, were very close to the *a priori* predictions based on this theories derivation chain in a recent large sample pre-registered study of ours in trained participants (Gschneider et al. 2025).

Whilst the theory that there may be a positive monotonic dose-response effect of fractional volumes upon muscle hypertrophy may in fact be true, it seems unlikely that any between-volume differences are of the magnitudes suggested in recent meta-regressions when examining previously trained participants. Researchers designing studies to act as strong tests of the theory of positive monotonic dose-response effects with RT volume in trained persons should also pay heed to the well corroborated theory of linear-log adaptation with duration of exposure to RT. As such, they should expect the magnitude of effects when comparing volumes to be small, even if real, in trained participants. Indeed, given the predicted main effects of time for RT, it seems unlikely that any difference between two RT interventions of different fractional volumes would even reach any reasonable smallest effect size of interest (SESOI). In our recent work we have begun to anticipate this and, following recent arguments for this in the sport and exercise sciences (Mazzolari et al., 2022), plan to test for equivalence within a defined SESOI (an SMD of [-0.1, 0.1]) instead of testing for the presence of an effect against a point null hypothesis of no difference (Gschneider et al. 2025).

Thus, whilst RT volume may indeed have a true effect upon muscle hypertrophy, this effect must necessarily be very small. As such, we sought to conduct a pre-registered test of the equivalence of two RT interventions of differing fractional weekly set volumes in previously trained participants. Our hypothesis was thus; in participants with at least 6 months of prior resistance training experience completing a 12 week intervention, the higher fractional volume resistance training condition will produce changes over time in our primary outcome measure of estimated muscle size (measured for both arm and thigh standardised and pooled across studies; see statistical analysis below) that are not larger (or smaller) than the SESOI when compared with

^aNote these are provided as the example as these are the volumes examined in the present study.

^bSee also the results of an arm-based meta-regression model examining the time-course of adaptation in the data from the aforementioned large meta-analysis (Steele et al. 2023) (see <https://osf.io/7fdvq>)

the lower fractional volume resistance training intervention condition i.e., will be statistically equivalent.

Methods

Experimental approach to the problem

A randomised controlled trial across multiple sites was employed. At the participant level, participants were randomly assigned to either the low volume or high volume RT intervention condition. Participants underwent baseline testing (t_0) and then participated in a 12-week intervention with post testing following the 12-week intervention (t_1). This study was pre-registered June 2025, available at: <https://doi.org/10.17605/OSF.IO/7VMXK>. This study was covered under prior ethical approval for by Solent University Health, Exercise, and Sports Science Research Ethics and Innovation Committee (reference number: fishj1HESS2024).

Updates to pre-registration

Note, our pre-registration underwent several updates which are documented clearly in the pre-registration due to both logistical reasons and some errors noticed in the original simulations and sample size planning. To summarise here for the reader, three major updates are relevant to highlight. Firstly, due to a clerical error some participants in the low volume group began the intervention period undertaking an earlier version of the proposed low volume intervention and as such it was decided to halt the study, allow a one-week washout of no training/testing, and then have them restart including baseline testing again. Secondly, due to an error in the original simulations for determination of sample size we realised that our original target for sample size would be insufficient to achieve acceptable power given the originally chosen and very conservative type 1 error rate ($\alpha = 0.01$). As such, after changing our analysis plan to the most powerful option from our simulations we also opted to increase our type 1 error rate to the more traditional rate utilised in the field ($\alpha = 0.05$) in order to enable us to achieve sufficient statistical power (i.e., ~80%). Lastly, as in our previous study (Gschneidner et al. 2025) we originally planned to utilise structured client workout records using StrengthPortal; however, prior to the study launch Discover Strength stopped using this software. As such, workout records were manually recorded across sites via googlesheet workout cards. This resulted in a large amount of unstructured workout data and, as this would have required a considerable amount of manual work to restructure, clean, and prepare this data, the decision was made to omit the secondary outcomes for strength as estimated 1RM from our analysis. Note, the methods described here pertain to the updated pre-registered methods.

Participants and Sample Size

Our pre-registered target sample size for recruitment was 120 participants (across 24 Discover Strength sites, target of 5 participants recruited at each site). The full details of the updated sample size justification for statistical power including simulations, assumptions, analyses, and inference criteria can be seen in the pre-registration. Briefly, we assumed the aforementioned theoretical linear-log growth function and estimated the expected main effect of time (SMD of ~0.05) for participants with the prior training experience required for inclusion, noted below, and assumed any condition by time interaction effect to be at most half of this (i.e., ~0.025). We considered this to be within what we felt to be the SESOI for changes in estimated muscle size; an SMD ranging [-0.1, 0.1]. We also corroborated the choice of this SESOI by consulting other researchers in the area and have utilised this in previous work (Gschneidner et al. 2025; Varovic et al. 2025). We then simulated to determine statistical power at different sample sizes assuming these effects, assuming a dropout rate of ~15% based on our previous studies (Carlson et al. 2023; Gschneidner et al. 2025), and testing for equivalence following the analysis plan detailed below.

A total of 125 participants were recruited from 22 locations of Discover Strength personal training studio (USA). A total of 120 participants were randomised to either the low ($n = 56$) or high volume conditions ($n = 64$). All participants were existing clients or staff with a minimum of 6 months RT experience with Discover Strength (though may have also had prior training experience before joining Discover Strength as

members) ensuring all participants were of a similar recent training background prior to participation. As such, all participants prior to beginning the study had been training for at least 6 months with supervised, high effort (i.e., training to momentary failure, and occasionally the use of advanced training techniques such as drop-sets, pre- or post-exhaustion, forced repetitions, rest-pause etc.), low-volume (i.e., a single set of each exercise), and twice-weekly training sessions. Participants were instructed not to (and confirmed with supervising trainers that they did not), engage in any muscle strengthening exercise outside of their supervised RT sessions. Participants were also asked to maintain normal dietary patterns and daily activities or other physical activities, exercise, and sports that they currently participated in (e.g., not to begin additional exercise strategies to enhance weight loss/gain). All participants signed an informed consent form prior to any data collection.

Muscle Area Measurement

The primary outcome was muscle hypertrophy operationalised as arm- and thigh- muscle area estimated from anthropometric measurements. Estimates of both arm muscle cross sectional area (CSA) and thigh muscle CSA were made from anthropometric measurements using methods and equations described previously by Heymsfield, et al. (1982) and Housh et al. (1995), respectively. For arm muscle CSA, circumference was measured at the midpoint between the tip of the acromion and the olecranon process with the arm hanging relaxed by the participant's side and taking the triceps skinfold as a vertical fold at the same point using callipers. For thigh muscle CSA, circumference was measured at the midpoint of the inguinal crease and the proximal border of the patella, and a thigh skinfold taken as a vertical fold at the same point using the same callipers. Measurements were taken by multiple instructors across multiple locations. All staff at each site conducting anthropometric measurements underwent initial training collecting measurements from at least 5 different people over two occasions. For each muscle CSA outcome three measurements were taken for both circumference and skinfold at each time point and all were used in analysis. This choice was based on simulations in planning our previous study (Gschneidner et al. 2025) regarding the impact of multiple measurements at each time point, whilst taking into account estimated measurement error determined from prior data, upon statistical power. Baseline test-retest data from our previous study (Gschneidner et al. 2025) estimated the standard error of measurement for estimated arm and thigh CSA to be 4.62 cm² and 7.97 cm² respectively.

Similarly to our previous study, our choice for these methods of operationalisation is pragmatic as, given the scale of the study being conducted across multiple sites, in order to reach sufficient sample size for statistical power we are limited in our ability to collect more direct measures such as ultrasound by the availability of equipment and trained personnel to collect that data. Indeed, Haun et al. (2019) have argued that, while such measurements might not offer microscopic insights into myofibrillar protein accrual or functional CSA, they are appropriate for examining changes due to RT interventions and particularly so when resource and technical constraints prevent the use of other methods. Further, we had previously reported (Gschneidner et al. 2025) a re-analysis of data for 67 participants (both untrained and trained) from a previous collection of studies where both circumference and ultrasound-based measures of hypertrophy had been used (Gentil et al. 2020). Calculating the pre- to post-intervention SMDs for both methods, and then comparing the methods using a random effects meta-analysis with method as a moderator revealed an SMD for circumference measures of 0.34 [95%CI: 0.23, 0.44], for ultrasound measures of 0.27 [95%CI: 0.19, 0.35], and difference between methods of -0.07 [95%CI: -0.2, 0.06]. Thus, both methods appear clearly able to detect changes in muscle size with relatively similar effect size magnitudes suggesting that the estimated CSA method used in this study are valid for their purpose.

Intervention

For both training intervention conditions (i.e., low and high volume) participants trained with a load permitting ~7-10 repetitions to be completed before reaching momentary failure (defined as the point at which, despite the greatest effort, the participant failed to complete the concentric phase of a repetition (Steele et al. 2017)) using a repetition duration of 2 second concentric and 4 second eccentric actions. The high volume condition had load altered in subsequent sets within the same session by the supervising trainer to enable an approximately similar repetition range to be achieved for each set. Both conditions performed three sessions

a week for 12 weeks. Unless specified the exercises were performed using resistance machines (combination of MedX, Imagine Strength Nautilus, and Hammer Strength depending on the available equipment at the site).

The high volume condition, in each session, performed the following workout of 12 fractional sets for the upper arm, and 12 fractional sets for the thigh, totalling 36 fractional sets for both upper arm and thigh per week:

- 3 sets of Leg Press (direct sets)
- 3 sets of Leg Curl (direct sets)
- 3 sets of Leg Extension (direct sets)
- 3 sets of Barbell/Dumbbell Romanian Deadlift (direct sets)
- 2 sets of Chest Press (indirect sets)
- 2 sets of Pulldown (indirect sets)
- 2 sets of Dumbbell Skullcrusher (direct sets)
- 2 sets of Dumbbell Split Stance Biceps Curl (direct sets)
- 2 sets of Overhead Press (indirect sets)
- 2 sets of Seated Row (indirect sets)
- 2 sets of Dumbbell French Press Tricep Extension (direct sets)
- 2 sets of Dumbbell Hammer Incline Curl (direct sets)

The low volume condition, in each session, performed one of the two following workouts in alternation consisting of 3 fractional sets for the upper arm, and 3 fractional sets for the thigh, totalling 9 fractional sets for both upper arm and thigh per week utilising the same core direct and indirect sets exercise selection as the high volume condition (in addition they completed a selection of exercises not directly or indirectly targeting the upper arm or thigh in order to increase the total session duration to be similar to typical session durations and accommodate the logistics of session timetabling at Discover Strength):

Workout A

- 1 set of Leg Press (direct set)
- 1 set of Leg Curl (direct set)
- 1 set of Leg Extension (direct set)
- 1 set of Tibia Dorsiflexion
- 1 set of Pec Fly
- 1 set of Pullover
- 1 set of Chest Press (indirect set)
- 1 set of Pulldown (indirect set)
- 1 set of Dumbbell French Press Tricep Extension (direct set)
- 1 set of Dumbbell Split Stance Biceps Curl (direct set)
- 1 set of Abdominal Flexion
- 1 set of Lumbar Extension

Workout B

- 1 set of Seated Calf Raise i.e., Plantarflexion
- 1 set of Leg Press (direct set)
- 1 set of Barbell/Dumbbell Romanian Deadlift (direct set)
- 1 set of Leg Extension (direct set)
- 1 set of Overhead Press (indirect set)
- 1 set of Seated Row (indirect set)
- 1 set of Shoulder Lateral Raise
- 1 set of Rear Shoulder Raise
- 1 set of Dumbbell French Press Tricep Extension (direct set)
- 1 set of Dumbbell Split Stance Biceps Curl (direct set)
- 1 set of Torso Rotation

The exercises selected were deemed to be either direct or indirect sets regarding fractional volumes based

upon the coding systems employed by Pelland et al. (2026) and Remmert et al. (2025) in their meta-regressions. Notably, our outcome measures pertained to the mid point of the upper arm and thigh as a whole, and so we considered this to include both elbow flexors/extensors, and knee flexors/extensors. Hence, we have considered exercises that are direct/indirect for either quadriceps, knee extensors, lateral thigh, anterior/middle thigh, vastus lateralis, vastus medialis, vastus intermedius, hamstrings or posterior thigh to be direct/indirect for the thigh as a whole, and exercises that are direct/indirect for either triceps brachii, elbow extensors, biceps brachii or elbow flexors to be direct/indirect for the upper arm as a whole. We corresponded with these authors to confirm that our approach reflected their coding sufficiently for our test of their volume model to be valid.

Statistical Analysis

All code utilized for data preparation and analyses are available in either the Open Science Framework page for this project <https://osf.io/bxa8p> or the corresponding GitHub repository https://github.com/jamessteeleii/project_DS_volume. We cite all software and packages used in the analysis pipeline using the `grateful` package (Rodriguez-Sanchez et al. 2023) which can be seen here: https://jamessteeleii.github.io/project_DS_volume/grateful-report.html.

As noted, the project was previously pre-registered including the analysis plan, model to be employed, parameter of primary interest, and specific hypothesis relating to this. The full details of this including the derivation of our hypotheses from prior evidence and theory regarding the expected effects of resistance training upon hypertrophy are fully detailed in the pre-registration for the reader (see <https://doi.org/10.17605/OSF.IO/7VMXX>). Here we reiterate our primary hypothesis related to the between condition comparison of low vs. high volume RT interventions upon muscle hypertrophy operationalised as arm and thigh estimated muscle CSA over time i.e., the time-by-condition interaction, where time is pre- and post-intervention (i.e., t_0 and t_1) and condition is the two aforementioned interventions. We tested for the equivalence of the slopes for time of high volume against the low volume comparator with a SESOI of 0.1 SMD. Our hypothesis is thus that the high volume resistance training condition will produce changes over time in our primary outcome of muscle hypertrophy that are not larger (or smaller) than the SESOI when compared with the low volume resistance training intervention condition. More specifically:

H0: The interaction effect for time (pre- and post-intervention) and condition (high fractional volume i.e., 36 fractional sets per week, or low fractional volume i.e., 9 fractional sets per week) for standardised (i.e., z-scored) estimated muscle area will differ from the smallest effect size of interest - upper and lower bound of confidence interval for between condition effect will be outside of or include the upper or lower limits of smallest effect size of interest i.e., [-0.1,0.1].

H1: The interaction effect for time (pre- and post-intervention) and condition (high fractional volume i.e., 36 fractional sets per week, or low fractional volume i.e., 9 fractional sets per week) for standardised (i.e., z-scored) estimated muscle area will be equivalent to the smallest effect size of interest - upper and lower bound of confidence interval for between condition effect will be inside the upper or lower limits of smallest effect size of interest i.e., [-0.1,0.1].

An analysis of covariance as a linear mixed effects location-scale model was fit using the `glmmTMB` package and using Restricted Maximum Likelihood estimation for pooled standardised arm and thigh estimated muscle CSA outcomes. We included fixed effects for time and condition:time interaction (our estimate of interest), random intercepts for both site id and participant id, for the location component (i.e., μ_i). The scale component (i.e., σ_i^2) of the model was to allow for heterogeneous variances between the two outcome measures, as they were modelled as pooled, and so included a fixed effect for outcome i.e., arm or thigh estimated muscle CSA. The model equation was as follows:

$$\begin{aligned}
y_i &\sim \mathcal{N}(\mu_i, \sigma_i^2) \\
\mu_i &= \beta_0 + \beta_1 (\text{time}_i) + \beta_2 (\text{time}_i \times \text{condition}_i) + u_{j[i]} + v_{k[i]} \\
u_j &\sim N(0, \sigma_{\text{participant}}^2) \quad \text{for participant } j = 1, \dots, J \\
v_k &\sim N(0, \sigma_{\text{site}}^2) \quad \text{for site } k = 1, \dots, K \\
\log(\sigma_i^2) &= \delta_0 + \delta_1 (\text{outcome}_i)
\end{aligned}$$

Where y_i was the estimated muscle CSA for either arm or thigh standardised following the approach of Penney (2023) i.e., using the error term of a simple linear model including only the randomised condition predictor avoiding possible attenuation issues with using z-scores^c. Condition was coded as centred (i.e., low = -0.5; high = 0.5) such that the main effect of time was interpretable as the mean effect across both conditions.

Our primary test for equivalence was upon the condition:time effect i.e., $\beta_2(\text{time}_i \times \text{condition}_i)$ from our primary pre-registered model using the `marginalEffects` packages `hypotheses()` function against our SESOI of 0.1. As noted, we originally pre-registered $\alpha = 0.01$ to draw inferences regarding equivalence, but this was updated to $\alpha = 0.05$ in order to maximise statistical power for our test. We also examined the estimated main time effect i.e., $\beta_1(\text{time})$ in our model descriptively in terms of it's magnitude and precision, and for visualisation present the un-pooled linear predictions for each participant on the raw scale (i.e., cm^2) in addition to the means and 95% quantile intervals for the raw pre- and post-intervention data.

Results

After randomisation a total of 112 participants underwent baseline testing (lower volume = 53; higher volume = 53) and 87 (lower volume = 42; higher volume = 45) participants completed post-testing. There was little difference in drop-out between conditions (low minus high volume = -5% [95%CI: -21%, 11%]; low volume drop-out = 25% [95%CI: 13%, 37%]; high volume drop-out = 30% [95%CI: 19%, 41%]).

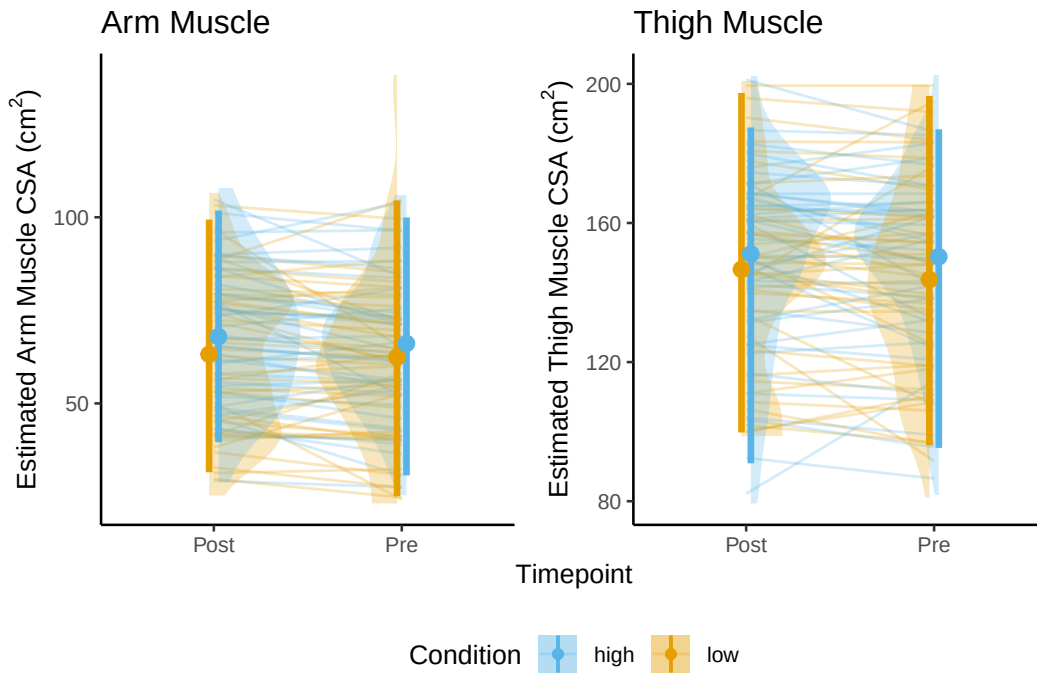
Our primary estimand of interest was the condition:time interaction effect from our pre-registered analysis reflecting the between condition difference in change in hypertrophy over time. The estimate for this effect was a standardised value of 0.023 [95%CI: -0.058, 0.104]. The p -value for equivalence was $p=0.032$. As such, considering our updated inference criteria with alpha set at 0.05, we were able to reject the null hypothesis that the condition:time interaction effect was outside of the SESOI [-0.1, 0.1] and thus claim statistical equivalence of effects. The raw estimated muscle CSA pre- and post-intervention predictions from un-pooled participant level linear regression in addition to the means and 95% quantile intervals can be see in Figure 1, along with the estimates for the main effect of time and condition:time interaction effect on the standardised scale from the pre-registered model.

Discussion

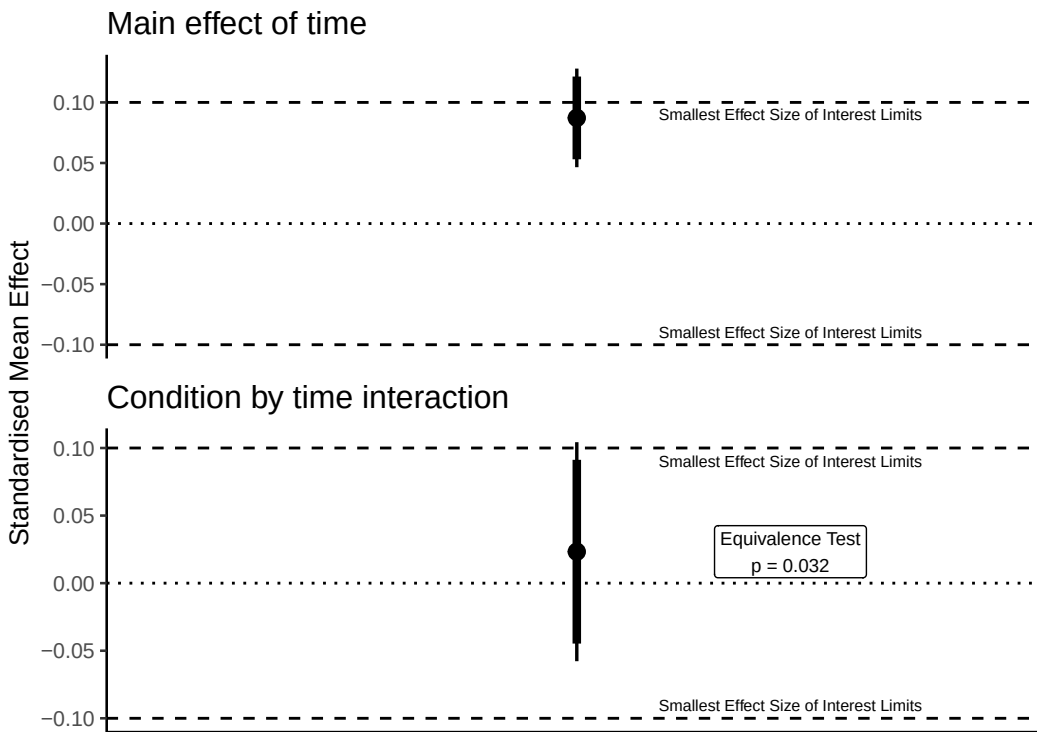
To our knowledge this is the largest study to have experimentally compared different RT volumes and offer a severe test (i.e., well powered, pre-registered) of the combined theories regarding the dose-response effects of RT volume and the linear-log adaptation to RT over time for hypertrophy. In a multi-site randomized trial we compared the effects of RT using low or high fractional weekly volumes upon muscle size in trained participants and tested for equivalence. The p -value for equivalence was 0.032 supporting that the effects of

^cNote, we deviate slightly from our pre-registered analysis here in that, instead of using the sample data to generate the denominator for standardising the estimated muscle CSA for either arm or thigh for analysis, we utilised the same denominators as estimated in our previous study (Gschneidner et al. 2025). The reasoning for this was that, given the nature of the present study, we had a comparatively smaller sample and so reasoned that the estimated σ from our previous and larger sample study was a better estimate. It would also render our results comparable with our previous study in terms of their exact magnitude. Notably, we realised in hindsight that we should have pre-registered this given that we anticipated more difficulty in recruitment and that we would not be able to obtain a sample size of the scale in our previous study given the nature of the interventions being examined in this population.

Primary Pre-registered Hypertrophy Outcomes



Raw means [95% quantile intervals] and participant level unpooled predictions



Thick error bars are 90% and thin error bars 95% confidence intervals

Figure 1: Primary pre-registered hypertrophy outcomes. The top two panels show the raw estimated muscle CSA pre- and post-intervention predictions from un-pooled participant level linear regression in addition to the means and 95% quantile intervals. The bottom two panels show estimates and 95% confidence intervals on the standardised scale from the pre-registered model for the main effect of time and the condition:time interaction effect.

the low and high volume conditions examined were statistically equivalent in trained participants (SMD = 0.023 [95%CI: -0.058, 0.104]).

The overall magnitude of RT effect in trained participants seen in the main effect of time in our models was 0.087 [95%CI: 0.047, 0.128] which was not dissimilar to the estimates reported in our previous work (0.034 [95%CI: 0.021, 0.046] and 0.051 [95%CI: 0.037, 0.064] for arm and thigh muscle respectively (Gschneidner et al. 2025)). This, combined with the statistical equivalence of the low and high volume conditions, further corroborate the theory that adaptation is diminished with continued participation in RT. Indeed, as this and our previous work show, the main effects of RT are likely far smaller than the typical studies in the field are designed to detect, let alone any between condition comparative effects.

Although we report statistical equivalence within the range of our SESOI in this study, it may remain a possibility that the theory of positive monotonic dose response effects of RT volume do indeed continue to play out in trained participants and perhaps it is just the case that the effects are very small. It could be argued that true small effects favouring higher volumes may yet still exist and that our study was unable to detect them. It is assuredly the case that our study would have been underpowered to detect very small effects and indeed alongside our simulations to determine power at different sample sizes when testing for equivalence we also examined power to test for the presence of a very small between condition effect i.e., ~ 0.025 . Such an effect would require >1000 participants to detect with even modest statistical power of ~ 60 -70% (see <https://osf.io/z8vck>). As such, whilst we cannot rule out that for a smaller SESOI we might not be able to infer statistical equivalence, it remains incumbent for those researchers who might want to claim that such small effects exist, and are important to detect, to conduct the appropriate studies to provide a strong test of that claim or estimate the effect with sufficient precision. From a practical perspective trainees and trainers might opt to take our results at face value and act as though both lower and higher volumes of the sort examined here are indeed equivalent in effects and then opt to train one way or the other based upon personal preference or other considerations. Alternatively, if they believe that there may still be greater benefits for higher volumes and do not see the additional time/resource commitment or other costs associated with such RT to be sufficient to dissuade them from engaging in it, then they may wish to pursue this kind of training for the possibility that it *may* provide them with additional hypertrophy. However, strong claims to the effect that higher volumes produce meaningfully greater hypertrophy than lower volumes in trained participants are currently unwarranted in our opinion.

The strengths of this study includes its sample size and pre-registration. Yet, despite this, we will pre-empt (and indeed reiterate given these have been highlighted already in our previous work; (Gschneidner et al. 2025)) some of the limitations that might be noted by readers. These include drop-outs and our chosen outcome operationalisations.

We noted the update to our pre-registration and inference criteria in order to maintain the desired statistical power. However, though in our previous studies with Discover Strength members we estimated a dropout rate of $\sim 15\%$ (Carlson et al. 2023; Gschneidner et al. 2025) and accounted for this in our simulations, the rate observed in the present study was almost double that. Fortunately there was not a clear difference between the low and high volume groups in terms of drop-outs and so we at least feel confident that differential drop-outs have likely not biased our estimates substantially. The drop-outs will have affected our statistical power and precision of our estimates, though even in light of this the present study remains one of the largest experimental trials of RT and the effects of volume. We speculate that the higher drop-out rate in this study may have been due to the increase in training frequency from the typical two sessions per week that Discover Strength usually employ to three sessions per week and the subsequent burden on participants.

Again, some may suggest that the operationalisations used for hypertrophy (i.e., estimated muscle CSA from circumference and skinfolds) in this study are not gold-standard and thus will have less resolution to detect effects due to greater measurement error. However, similarly to our previous work, we acknowledged this in study planning and explicitly estimated the measurement error with our prior data using these methods, used this in simulating to determine sample size for the desired statistical power, and also made use of multiple measurements to account for the error associated with single measurements. Additionally, where in our previous work we modelled arm and thigh outcomes separately and opted to adjust our alpha levels given our global hypothesis regarding “hypertrophy” as an outcome, here we pooled standardised outcomes and

modelled these using an appropriate location-scale model allowing for heterogeneous variances and providing greater statistical power. Again, we feel confident that our operationalisations reflect the construct of interest i.e., hypertrophy (Haun et al. 2019), given the striking closeness of our estimates to the *a priori* predictions regarding the effects; predictions which were derived from both theory and empirical evidence which had employed a wide range of operationalisations for both constructs.

Conclusion

This study is, to our knowledge, one of the largest to compare the effects of low and high volume RT interventions upon hypertrophy in previously trained participants. We found statistical equivalence between both conditions and both main effects of time, and any interaction effects for condition by time, are likely small. More broadly, this study further corroborates the linear-log theory of adaptation, that the effects of RT in trained persons should be expected to be small, and that current studies in the field of RT are woefully underpowered to be able to detect their effects, let alone test between intervention comparisons.

Acknowledgements

We would like to extend our sincere gratitude to all the Exercise Physiologists at Discover Strength without whom this project would not have been possible: Downtown Minneapolis, MN: Chris Langley, Brandon Schuchard, Bobby Davis III, Darina Cerza. Northeast Minneapolis, MN: Maria Hauger, Lawrence Reed, Edwin Olmedo, Allie Rohwer, Sophie Bartell, Teresa Winkelman, Chris Nelson. Chicago River North, IL: Kyle Snuttjer, Luke Shepherd. Scottsdale, AZ: Megan Palmer, Mitchell McIntyre, Taylor Triplett. St. Louis Park, MN: Grant Whitton, Zachary Leris, Levi Wolter, Taylor Melvin, Colby Deuth, Tristan Masuga, Lucas Hoyt, Graham Proulx. Plymouth, MN: Rick Rignell, Kwaku Akore, Kayla Ezuck, Kyle McKinney, Alex Cizek, Jake Petersen, Micah Madyun, Rachel Holmes. Woodbury, MN: Ben Richter, John McQuade, Chase Thomley, Matt Xiong, Sven Otterness, Madison Allie, Tobi Snow, Cole Truesdell. Draper, UT: Alessandra Wendt, Jeff Wendt, Parker Hewes, Sophie Hamling. Maple Grove, MN: Jim Trnka, Sam Hoppenrath, Taylor Rachjel, Luke Billehus, Colton Haight, Cullen Stamp, Brad Stanke, Blake Boggs. Apple Valley, MN: Caleb Countryman, Abe Allen, Tyler Barry, Maddie Sarvela. Wayzata, MN: Aaron Zambrano-Vega, Curtis Hoang, Andy Alessio, John Unverzagt, Ari Smolen, Keyan Mayginnnes, Tristan Peter, Tyler Gustafson. Chanhassen, MN: Austin Carter, Kristina Harder, Tommy Picka, Logan Morrison, Andrew Kittelsen, Isaac Erickson-Thoemke, Gabe Petersen, Alex Gallen, Lexi Kolb, Isaiah Cherrier. Rochester, MN: Darby Johnson, Tanner Jensen. Edina, MN: Tyler Zolczynski, Mason Roeglin, Alexis Jones, Blake Gulden, Grace Vaughan-Wenner, Derek Lauson, Anna Spencer, Kaitlyn Zignego, Nick Katzman, Jack Schneider, Angel Denis, Maddie Ehrich-Wilking. Savage, MN: Titus Malika, Justin Clark.

Further, we would like to thank all participants in this research study for their engagement in this project and their commitment to their long-term health.

Funding Information

No direct funding was received for this study. Operational costs for its delivery were absorbed by Discover Strength and so provided *in-kind* value.

Author Contributions

JS, DG, LC, and JF conceived the idea for the project and designed the methods. LC and DG coordinated and oversaw the data collection. JS conducted the statistical analyses and produced data visualisations. JS drafted the initial manuscript. JS, DG, LC, and JF contributed to editing the manuscript. All authors read and approved the final manuscript.

Competing Interests

LC and DG work for Discover Strength. JF provides research consulting for organizations within the health and fitness field and has received travel expenses and honoraria for speaking and consulting with Discover Strength. JS provides research consultancy through his company Steele Research Limited, is contracted currently by MacroFactor and Kieser Australia through Steele Research Limited, and has also received travel expenses and honoraria for speaking from fit20 International, Exercise School Portugal, and Discover Strength.

The content does not necessarily represent the official views of the affiliations or contracting organisations listed.

Data and Supplementary Material Accessibility

All data and code utilised for data preparation and analyses are available in either the Open Science Framework page for this project <https://osf.io/bxa8p/overview> or the corresponding GitHub repository https://github.com/jamessteelei/project_DS_volume.

References

- Carlson, Luke, David Gschneidner, James Steele, and James P. Fisher. 2023. "The Effects of Training Load During Dietary Intervention Upon Fat Loss: A Randomized Crossover Trial." *Research Quarterly for Exercise and Sport* 94 (4): 990–1000. <https://doi.org/10.1080/02701367.2022.2097625>.
- Gentil, Paulo, Emily Budzynski-Seymour, Daniel Souza, James Steele, James P. Fisher, and Martim Bottaro. 2020. "Evaluating the Results of Resistance Training Using Ultrasound or Flexed Arm Circumference: A Case for Keeping It Simple?" *Journal of Clinical and Translational Research* 7 (6): 61–65.
- Gschneidner, David, Luke Carlson, James Steele, and James P. Fisher. 2025. "The Effects of Lengthened-Partial Range of Motion Resistance Training of the Limbs on Arm and Thigh Muscle Area: A Multi-Site Randomised Trial." *Journal of Sports Sciences* 43 (23): 2963–76. <https://doi.org/10.1080/02640414.2025.2567805>.
- Haun, Cody T., Christopher G. Vann, Brandon M. Roberts, Andrew D. Vigotsky, Brad J. Schoenfeld, and Michael D. Roberts. 2019. "A Critical Evaluation of the Biological Construct Skeletal Muscle Hypertrophy: Size Matters but So Does the Measurement." *Frontiers in Physiology* 10 (March). <https://doi.org/10.3389/fphys.2019.00247>.
- Heymsfield, S. B., C. McManus, J. Smith, V. Stevens, and D. W. Nixon. 1982. "Anthropometric Measurement of Muscle Mass: Revised Equations for Calculating Bone-Free Arm Muscle Area." *The American Journal of Clinical Nutrition* 36 (4): 680–90. <https://doi.org/10.1093/ajcn/36.4.680>.
- Housh, D. J., T. J. Housh, J. P. Weir, L. L. Weir, G. O. Johnson, and J. R. Stout. 1995. "Anthropometric Estimation of Thigh Muscle Cross-Sectional Area." *Medicine and Science in Sports and Exercise* 27 (5): 784–91.
- Latella, Christopher, Daniel van den Hoek, Milo Wolf, Patroklos Androulakis-Korakakis, James P. Fisher, and James Steele. 2024. "Using Powerlifting Athletes to Determine Strength Adaptations Across Ages in Males and Females: A Longitudinal Growth Modelling Approach." *Sports Medicine (Auckland, N.Z.)* 54 (3): 753–74. <https://doi.org/10.1007/s40279-023-01962-6>.
- Pelland, Joshua C., Jacob F. Remmert, Zac P. Robinson, Seth R. Hinson, and Michael C. Zourdos. 2026. "The Resistance Training Dose Response: Meta-Regressions Exploring the Effects of Weekly Volume and Frequency on Muscle Hypertrophy and Strength Gains." *Sports Medicine* 56 (2): 481–505. <https://doi.org/10.1007/s40279-025-02344-w>.
- Penney, Jeffrey. 2023. "Cautions When Normalizing the Dependent Variable in a Regression as a z-Score." *Economic Inquiry* 61 (2): 402–12. <https://doi.org/10.1111/ecin.13127>.
- Remmert, Jacob, Joshua Pelland, Zac Robinson, Seth Hinson, and Michael Zourdos. 2025. "Is There Too Much of a Good Thing? Meta-Regressions of the Effect of Per-Session Volume on Hypertrophy and Strength." SportRxiv. <https://doi.org/10.51224/SRXIV.537>.

- Rodriguez-Sanchez, Francisco, cre, cph, Connor P. Jackson, Shaurita D. Hutchins, and James M. Clawson. 2023. “Grateful: Facilitate Citation of R Packages.”
- Steele, James, James P. Fisher, Jurgen Giessing, Patroklos Androulakis-Korakakis, Milo Wolf, Bram Kroeske, and Rob Reuters. 2022. “Long-Term Time-Course of Strength Adaptation to Minimal Dose Resistance Training Through Retrospective Longitudinal Growth Modeling.” *Research Quarterly for Exercise and Sport* 0 (0): 1–18. <https://doi.org/10.1080/02701367.2022.2070592>.
- Steele, James, James Fisher, Jürgen Giessing, and Paulo Gentil. 2017. “Clarity in Reporting Terminology and Definitions of Set Endpoints in Resistance Training.” *Muscle & Nerve* 56 (3): 368–74. <https://doi.org/10.1002/mus.25557>.
- Steele, James, Fisher, Smith, Schoenfeld, Yang, and Shinichi and Nakagawa. 2023. “Meta-Analysis of Variation in Sport and Exercise Science: Examples of Application Within Resistance Training Research.” *Journal of Sports Sciences* 41 (17): 1617–34. <https://doi.org/10.1080/02640414.2023.2286748>.
- Varovic, Dorian, Milo Wolf, Brad J. Schoenfeld, James Steele, Jozo Grgic, and Pavle Mikulic. 2025. “Does Muscle Length Influence Regional Hypertrophy? A Systematic Review and Meta-Analysis.” *International Journal of Sports Medicine* 46 (14): 1027–36. <https://doi.org/10.1055/a-2615-4935>.